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Data-driven prediction of striker productivity using Random Forest regression: Insights for enhancing Indonesian football performance

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Abstract

This study uses Random Forest Regression to identify key indicators affecting goal-scoring among forwards, with relevance to Indonesian football. Results show strong links between goals and shooting efficiency (G/Sh, G/SoT), playing time, and shot accuracy. The model estimates optimal values to boost output, offering data-driven insights for improving striker performance. These findings are valuable for teams like Indonesia, which lack access to advanced data. By turning stats into actionable goals, this research aids evidence-based training. Future work should include tactical, spatial, and seasonal data to enhance accuracy and support real-time scouting.

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1. Introduction

Football has long been a sport that captivates Indonesian society, with its origins rooted in the colonial era and its ongoing evolution into an integral component of the national culture [1]. Public enthusiasm is clearly manifested by the substantial support for local competitions, such as BRI Liga 1, and the considerable attention directed toward international events such as the FIFA World Cup [2]. Numerous football academies have been established in Indonesia, with the expectation that they will produce local professional football players [1, 3]. Nonetheless, within the context of national team development, significant challenges persist in maximizing the potential of local players, particularly

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in the attacking or striker positions [4, 5]. One of the striking indicators is the reliance on naturalized players [6], which reflects the suboptimal system for identifying and nurturing local talent, particularly in attacking positions.

This study arises from the need to identify the best attackers, defined as players capable of scoring a high number of goals. Its primary focus is on exploring the key factors influencing goal-scoring performance, such as shot accuracy (i.e., shots on target), the ratio of shots to goals, and playing time contribution. An understanding of these factors is essential for developing a data-driven and objective training system.

To this end, this study develops a predictive model based on the Random Forest Regression algorithm, leveraging the "2022–2023 Football Player Stats" dataset from Kaggle [7]. The data are sourced from Football Reference, a website that provides player statistics from European and American football leagues. Random Forest Regression was chosen due to its capability to handle complex variables and to offer insights into the significance of each feature in the prediction process [8–10]. Previous research has demonstrated that Random Forest can achieve high accuracy in predicting both individual performance and match outcomes across various sports, including football [11–13].

The Random Forest Regression model has also proven effective in modeling performance based on specific metrics such as expected goals (xG), shots, and passing patterns [14], even accurately predicting scores in major tournaments such as the FIFA World Cup [15]. Conversely, a study by Luo et al. demonstrated that combining Random Forest with team play structure information yields a more precise predictive model for final match scores [16].

After building the model, the next step in this study is to employ it to explore the optimal value of each key factor in order to maximize goal scoring. This aspect has been scarcely explored in previous studies, which have primarily focused on performance classification or team outcome simulations rather than on regressing individual performance metrics such as goal count. By identifying the most influential factors and their optimal values, coaches and the management of the Indonesian national team can design more targeted and evidence-based training programs.

Thus, this research not only contributes theoretically to the field of sports analytics but also provides a strategic foundation for enhancing the quality of local players, particularly strikers, in efforts to strengthen the autonomy and competitiveness of the Indonesian national team.

2. Literature Review

Random Forest is one of the ensemble-based machine learning algorithms widely utilized for its ability to generate accurate predictions and its resilience against overfitting. The algorithm operates by constructing multiple decision trees from random subsets of the training data and then combining their outcomes through a voting process for classification or averaging for regression. In the context of regression, Random Forest Regression is used to predict continuous values—such as the number of goals—by calculating the average prediction from all the regression trees. Its primary advantages lie in its flexibility to handle both numerical and categorical variables, as well as its capacity to estimate the importance of each variable within the model.

In recent years, several machine learning-based studies have explored how physical, technical, and tactical attributes can be leveraged to predict player performance. The application of Random Forest in the sports domain has yielded robust and varied results. For instance, in a study by Morciano et al., the Random Forest model was employed to predict football player performance based on biometric data collected from sensors during training sessions, achieving prediction accuracies exceeding 90% [17]. Similarly, Merzah et al. applied this algorithm to classify player performance into three categories based on match and training data [18]. In the context of scouting and recruitment, Ghar et al. developed a simulation system based on Random Forest to evaluate the suitability of new players for various team tactics, emphasizing the importance of aligning playing styles during the selection process [19]. Additionally, Ćwiklinski et al. adopted a regression approach to assess the success of player transfers using historical data, demonstrating a strong correlation between statistical indicators and transfer outcomes [20].

Regression approaches using Random Forest have also been employed to predict batsman performance in cricket [21], weightlifting success [22], and even the accuracy of attendance predictions for NBA games [23]. Sanjaykumar et al. utilized Random Forest to forecast match outcomes in the IPL cricket league with exceptionally high predictive accuracy, underscoring the model's relevance for team strategy and performance management [24]. In the realm of global sports, Schauberger and Groll demonstrated the superiority of Random Forest in predicting FIFA World Cup

match outcomes by modeling the number of goals as the target variable [15]. Moreover, in non-sport contexts, this model has consistently exhibited high performance, as seen in the prediction of student academic performance [25] and Olympic results [26], thereby illustrating the extensive applicability and adaptability of this approach.

Despite numerous studies leveraging Random Forest Regression in sports performance contexts, most research has focused on predictions based on physical or general historical features, without specifically addressing the quantitative relationships between technical attributes of football players—such as shots on target, shot-to-goal ratios, and minutes played—and the number of goals scored. Furthermore, no study has systematically explored the optimal values of these variables for maximizing goal productivity, particularly for attacking positions. Additionally, the majority of prior research has been confined to major European or American leagues, leaving a gap in models applicable to the development of local players in developing countries like Indonesia, where performance data infrastructure is limited. Lastly, predictive validation involving local football experts as model evaluators has been infrequently conducted, despite its importance in ensuring the practical relevance of the model outcomes.

This study aims to bridge these gaps by developing a Random Forest Regression model capable of predicting players' goal counts based on statistical data from the open "2022–2023 Football Player Stats" dataset. In addition to identifying the most influential variables affecting goal-scoring productivity, the research also explores the optimal values for each key variable, which can serve as a reference for training strikers for the Indonesian national team. Consequently, the findings are expected not only to contribute academically to player performance modeling but also to provide a practical and context-specific analytical tool for decision-makers in national football.

3. Methodology

This study began with the collection of data and its subsequent preparation. This was followed by the development of a machine learning model using Random Forest Regression. Next, the model's predictive performance was tested, and an analysis of the results provided by the model was conducted. An illustration of the research methodology is presented in Fig. 1.

3.1. Data Collecting and Preprocessing

	Rk	Player	Nation	Pos	Squad	Comp	Age	Born	MP	Starts	...	Off	Crs	TklW	PKwon	PKcon	OG	Recov	AerWon	AerLost	AerWon%
0	1	Brenden Aaronson	USA	MFFW	Leeds United	Premier League	22	2000	20	19	...	0.17	2.54	0.51	0.0	0.0	0.00	4.86	0.34	1.19	22.2
1	2	Yunis Abdelhamid	MAR	DF	Reims	Ligue 1	35	1987	22	22	...	0.05	0.18	1.59	0.0	0.0	0.00	6.64	2.18	1.23	64.0
2	3	Himad Abdelli	FRA	MFFW	Angers	Ligue 1	23	1999	14	8	...	0.00	1.05	1.40	0.0	0.0	0.00	8.14	0.93	1.05	47.1
3	4	Salis Abdul Samed	GHA	MF	Lens	Ligue 1	22	2000	20	20	...	0.00	0.35	0.80	0.0	0.0	0.05	6.60	0.50	0.50	50.0
4	5	Laurent Abergel	FRA	MF	Lorient	Ligue 1	30	1993	15	15	...	0.00	0.23	2.02	0.0	0.0	0.00	6.51	0.31	0.39	44.4

5 rows × 124 columns

Fig. 1. Original Data

The data used in this study were obtained from the "2022-2023 Football Player Stats" dataset on Kaggle, compiled by Vivo Vinco and licensed under the Attribution 4.0 International (CC BY 4.0) license. This dataset encompasses the physical, technical, and tactical attributes of football players worldwide, rendering it both relevant and well-structured. Fig. 1 illustrates the initial state of the data, which comprised over 2,500 rows and 124 columns. After data collection, a series of preprocessing steps were undertaken, including:

- Filtering: The dataset was filtered to include only entries with the **Pos** values **FW**, **MFFW**, **FWMF**, and **DFFW**, corresponding to forward line positions. This filtering process reduced the dataset to 898 rows.
- Data Cleaning: Missing values and anomalies were either removed or corrected.
- Data Correlation: The correlation between each feature and the **Goal** variable (the number of goals scored) was examined. Ten features were identified as having high correlations with the **Goal** variable. G/SoT, SoT, and SoT excluding penalty kicks:

- **Min** (minutes played) : 0.71
- **90s** (minutes played divided by 90) : 0.71
- **Starts** (matches started) : 0.70
- **MP** (matches played) : 0.57
- **G/SoT** (goals per shot on target) : 0.50
- **G/Sh** (goals per shot) : 0.47
- **ShoPK** (penalty kicks taken) : 0.42
- **PKatt** (penalty kick attempts) : 0.41
- **SoT** (shots on target) : 0.32
- **SoT%** (percentage of shots on target) : 0.28

Fig. 2 presents the current state of the data, which now consists of 898 rows and 10 columns. The dataset will subsequently be partitioned into two segments: the Goal column as the target variable (y) and the remaining nine columns as predictor variables (x).

	Goals	Min	90s	Starts	MP	G/SoT	G/Sh	ShoPK	PKatt	SoT	SoT%
0	1	1596	17.7	19	20	0.20	0.04	0.0	0.0	0.28	18.5
2	0	770	8.6	8	14	0.00	0.00	0.0	0.0	0.35	33.3
6	1	185	2.1	2	5	0.33	0.11	0.0	0.0	1.43	33.3
7	1	107	1.2	0	11	0.50	0.17	0.0	0.0	1.67	33.3
9	5	1507	16.7	17	22	0.29	0.11	0.0	0.0	1.02	37.0

Fig. 2 Dataset Result from Preprocessing Data

3.2. Machine Learning Model Development

Based on the literature review and the processed data, this study employs the Random Forest algorithm. This algorithm operates by constructing numerous decision trees at random and then aggregating their outputs through voting (for classification) or averaging (for regression) to enhance accuracy and reduce overfitting. Its primary advantages lie in its capability to handle both numerical and categorical data and its ability to provide insights into the importance of specific features.

The dataset was subsequently divided into two portions: 80% for training (718 samples) and 20% for testing (180 samples). To determine the optimal parameters, GridSearch was utilized, yielding the following optimal settings: max_depth = 10; min_samples_leaf = 1; min_samples_split = 2; n_estimators = 200; and random_state = 42.

To evaluate the model's performance on the test data, two evaluation metrics were employed, namely Mean Squared Error (MSE) and R-Squared. The results demonstrated an MSE of 0.247, indicating a low level of error, and an R-Squared of 0.96, which suggests that the model is capable of explaining approximately 97% of the variance in the target variable.

The subsequent step involved determining feature importance. The analysis of the variables with the highest importance revealed those features that exert the greatest influence on predicting the target variable. As shown in Fig. 3, the top five features identified were G/Sh, Min, Starts, SoT, and G/SoT. This refined dataset will then be used to explore and analyze the optimal values for each key factor, with the objective of maximizing the number of goals scored.

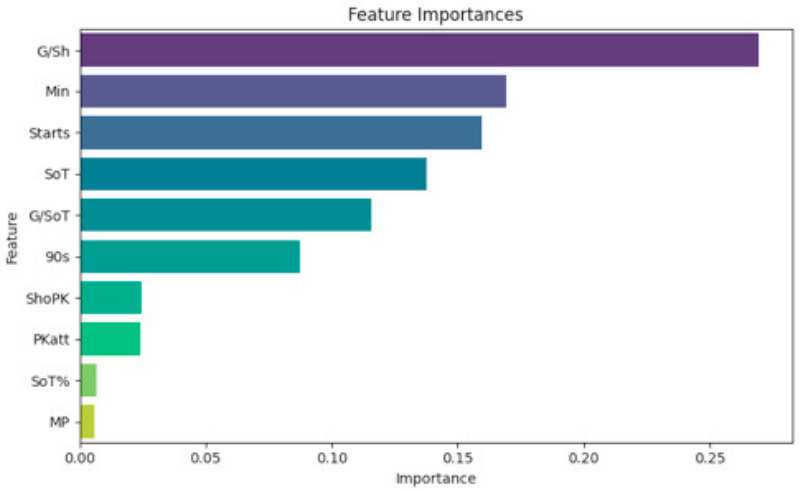


Fig. 3 Feature Importance Result

4. Result and Discussion

The subsequent step involves feature engineering by creating a new feature, "**Goal Category**," to classify the number of goals. Given that the minimum number of goals is 0 and the maximum is 25, several categories are defined as follows: 0 = 0 goals ; 1–5 = 1–5 goals; 6–10 = 6–10 goals ; 11–15 = 11–15 goals; 16–20 = 16–20 goals; 21–25 = 21–25 goals . The outcome of this new feature construction is illustrated in Fig 4. Subsequently, separate datasets will be created based on these categories to determine the minimum and maximum values of each feature that contribute to the highest goal output.

	G/Sh	Min	Starts	SoT	G/SoT	Goals	Goal Category
0	0.04	1596	19	0.28	0.20	1	1-5
2	0.00	770	8	0.35	0.00	0	0
6	0.11	185	2	1.43	0.33	1	1-5
7	0.17	107	0	1.67	0.50	1	1-5
9	0.11	1507	17	1.02	0.29	5	1-5

Fig. 4 Current dataset with new feature: Goal Category

An overview of these performance groups based on the metrics G/Sh, Min, Starts, SoT, G/SoT, and Goals is presented in Table 1. A clear progression in offensive contribution is evident across the groups. In Group 1 (0 goals), players exhibited minimal match involvement, characterized by low minutes played, limited starts, and no recorded shots on target or goals. Group 2 (1–5 goals) showed moderate improvement, with increased minutes, starts, and a modest rise in SoT and goal counts, indicating basic offensive involvement.

Table 1. Final Results

index	Goal Category	G/Sh (mean)	Min (mean)	Starts (mean)	SoT (mean)	G/SoT (mean)	Goals (mean)
0	Group 1 (0)	0.00	226.55	2.17	0.42	0.0	0.0
1	Group 2 (1-5)	0.14	814.55	9.01	0.93	0.37	2.32
2	Group 3 (6-10)	0.18	1359.46	16.10	1.13	0.42	7.40
3	Group 4 (11-15)	0.21	1542.89	17.89	1.34	0.46	12.39
4	Group 5 (16-20)	0.21	1728.00	19.50	1.69	0.50	16.50

5	Group 6 (21-25)	0.30	1636.00	19.00	1.98	0.58	25.00

In Group 3 (6–10 goals), the data showed a significant increase in playing time, starts, and offensive involvement, with notable improvements in SoT and goal production. Players in this group assumed a more active attacking role. Group 4 (11–15 goals) and above displayed progressively higher offensive efficiency, particularly in G/Sh and G/SoT, aligning with studies that highlight the critical role of shooting precision and minutes played in striker productivity [17]. The consistent upward trajectory in minutes played, SoT, and especially in efficiency metrics across groups highlights the relationship between match exposure, opportunity creation, and finishing proficiency. Higher minutes enable forwards to develop in-game rhythm and tactical synchronization [27], while rising G/Sh and G/SoT values indicate a shift from shot volume to shot precision — a hallmark of modern attacking football [28]. For coaches, this suggests Indonesian strikers need not just more minutes, but training focused on improving SoT and conversion under pressure. For scouts, SoT > 1.5 and G/SoT > 0.5 can serve as reliable indicators of high striker potential, consistent with international benchmarks [29].

To assess local applicability, a dataset of 25 Indonesian national team forwards was compiled in Table 2, featuring identical performance metrics. Players were classified into the Goal Categories, revealing that 52% of players were in the Group 1, while 48% fell into the Group 2. None achieved more than 5 goals.

Table 2. Goal Category Distribution – Indonesian Natioanl Team Forwards [30]

No	Player Name	G/Sh	Min	Starts	SoT	G/SoT	Goals	Goal Cat.
1	Rafael Struick	0,2	899	11	4	0,25	1	Group 2
2	Hokky Caraka	0	238	4	0	0	2	Group 2
3	Dimas Drajad	0	135	2	0	0	3	Group 2
4	Ramadhan Sananta	3	119	0	0	0	4	Group 2
5	Septian Bagaskara	0	0	0	0	0	0	Group 2
6	**Ole Romeny**	**0,25**	**354**	**4**	**6**	**0,33**	**3**	**Group 1**
7	Ragnar Oratmangoen	0,17	859	10	2	1	2	Group 2
8	Marselino Ferdinan	0,17	838	8	5	0,4	2	Group 2
9	Witan Sulaeman	0	540	5	1	0	1	Group 2
10	Ricky Kambuaya	0	388	4	0	0	0	Group 2
11	**Dendy Sulistyawan**	**0**	**180**	**3**	**0**	**0**	**0**	**Group 1**
12	**Egy Maulana**	**2**	**264**	**2**	**0**	**0**	**2**	**Group 1**
13	Saddil Ramdani	0	237	2	0	0	1	Group 2
14	Thom Haye	0,33	900	11	2	1	2	Group 2
15	Ivar Jenner	0	608	6	1	0	0	Group 2
16	**Joey Pelupessy**	**0**	**238**	**3**	**0**	**0**	**0**	**Group 1**
17	**Rachmat Irianto**	**0**	**180**	**2**	**0**	**0**	**0**	**Group 1**
18	**Marc Klok**	**0**	**120**	**2**	**0**	**0**	**0**	**Group 1**
19	**Yance Sayuri**	**0**	**90**	**1**	**0**	**0**	**0**	**Group 1**
20	**Malik Risaldi**	**0**	**63**	**1**	**0**	**0**	**0**	**Group 1**
21	**Beckham Putra**	**0**	**62**	**1**	**0**	**0**	**0**	**Group 1**
22	**Adam Alis**	**0**	**45**	**1**	**0**	**0**	**0**	**Group 1**
23	**Arkhan Fikri**	**0**	**64**	**0**	**0**	**0**	**0**	**Group 1**
24	**Stefano Lilipaly**	**0**	**40**	**0**	**0**	**0**	**0**	**Group 1**
25	**Eliano Reijnders**	**0**	**108**	**0**	**1**	**0**	**0**	**Group 1**

This validation confirms the model's applicability, highlighting that most Indonesian strikers currently occupy the lower performance categories. It reinforces recommendations that local forwards need structured development in shooting efficiency and tactical integration to elevate performance toward international standards. From a coaching perspective, these insights suggest that development programs for Indonesian forwards should not only increase minutes played but also implement targeted training focused on shot accuracy under match-like pressure conditions, and decision-making in optimal shooting zones. For scouts and analysts, this study provides practical thresholds

derived from high-performing groups — for example, forwards with SoT > 1.5 and G/SoT > 0.5 — which could serve as evidence-based benchmarks when evaluating striker prospects. Additionally, match strategies should emphasize creating high-probability scoring opportunities rather than focusing solely on shot quantity, optimizing player roles and decision-making pathways in the attacking third. These insights directly support national team coaches and scouts in refining selection criteria and individual development targets for future striker prospects.

5. Conclusion and Further Study

These findings are highly relevant to the primary objective of the study—to identify the key factors influencing the number of goals scored by forwards. It can be concluded that, in addition to the quantity of playing minutes and active participation on the pitch, the efficiency in converting scoring opportunities is a crucial aspect that players need to enhance. This provides concrete recommendations for the development programs of the Indonesian national team's forwards, suggesting that training should focus not only on increasing shot frequency but also on improving shot accuracy and the conversion of shots on target into goals. Furthermore, the results indicate that the Random Forest Regression model developed in this research can serve not only as a predictive tool but also as an analytical instrument for setting individual player performance targets. For example, "to score more than 10 goals per season, an ideal forward should have G/Sh > 0.2, SoT > 1.3, and playing minutes > 1500." Such information is highly valuable for coaches and analysts when designing data-driven training strategies.

This study also opens several avenues for further exploration that could broaden the scope and deepen the understanding of the key factors in forward performance. One potential direction is the integration of tactical and spatial variables—such as shooting zones, types of assists, and involvement in build-up play—which are not covered in the current model due to data limitations. Additionally, using longitudinal data from multiple seasons would enable a dynamic analysis of player performance trends, including the potential for predicting injuries or performance declines. Future research could also compare the effectiveness of the Random Forest algorithm with other approaches, such as XGBoost, Gradient Boosting Machines, or deep learning models, to determine the most optimal method for predicting goal counts. Finally, validating the model in the context of Indonesia's local league is essential to ensure that the results and recommendations are truly applicable to the domestic competition system and playing style. By broadening the range of variables and strengthening local validation, future studies can provide a more robust foundation for developing data-driven scouting and player development systems in Indonesia.

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